

Concept Review

Passive Infrared Sensors

Why Do We Need Passive Infrared Sensors?

Ever entered a room and have lights turn on automatically? Or ever play a game that detects when you're stationary versus in motion? To accomplish these tasks electromechanical systems can make use of passive infrared sensors (PIR) to detect changes in infrared (heat) energy. PIR sensors fall within the family of motion sensors and can be used in a wide range of applications.

Passive Infrared Sensors

Everything radiates some amount of heat in the form of infrared light, which is invisible to human eyes but detectable through infrared sensors. Passive infrared (PIR) sensors utilize piezoelectric ceramic materials to detect changes in infrared emission.

Typically, PIR sensors combine both a sensor plat and a lens to focus the infrared emissions. Figure 1 is an example of a possible sensor and lens combination for the Kemet PL- Q873.



Figure 1: electromechanical elements of the Kemet PL-Q873 PIR sensor

PIR sensors are most often used to detect motion. To accomplish this, the simplest form of a PIR sensor consists of two IR measuring elements, shown in Figure 2. In an idle environment, the two elements measure a similar amount of IR. When an object moves across the detection zones, a differential signal is generated between them, producing a spike in the voltage signal.

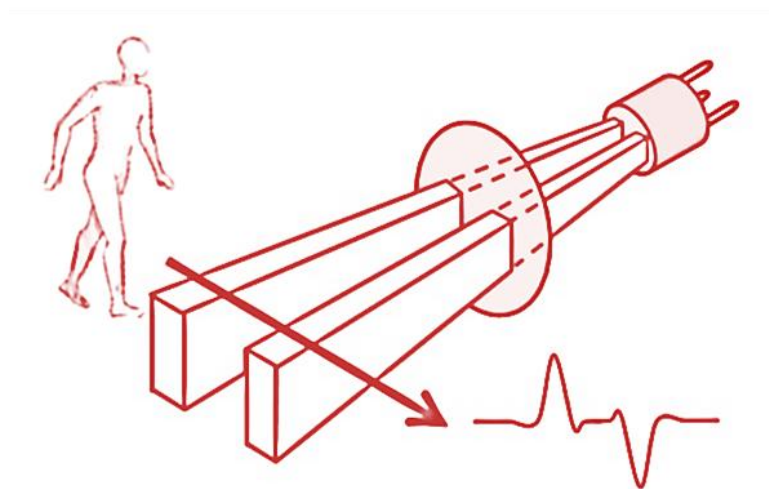


Figure 2: PIR sensors detect differential infrared levels. Figure adapted from:
https://www.allaboutcircuits.com/uploads/articles/STMicro1_.jpg

Many PIR sensors are used with a lens, which can also be seen in Figure 2, to help focus incoming IR light onto the sensing elements. The lens is a critical component that determines the sensor's key specifications of detection range and field of view.

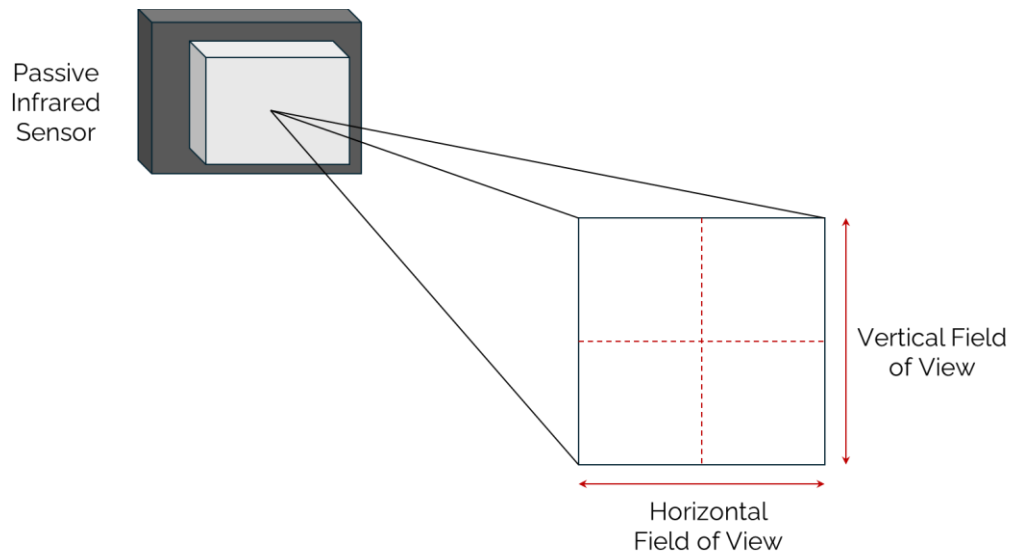
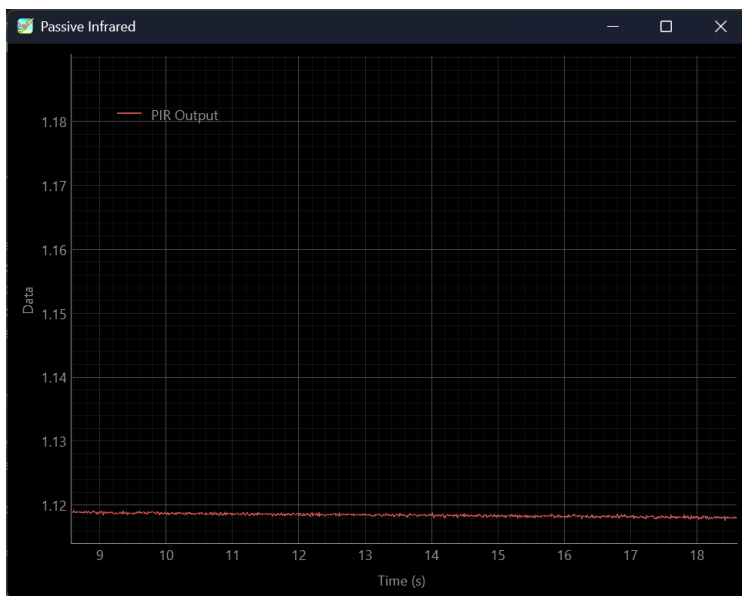
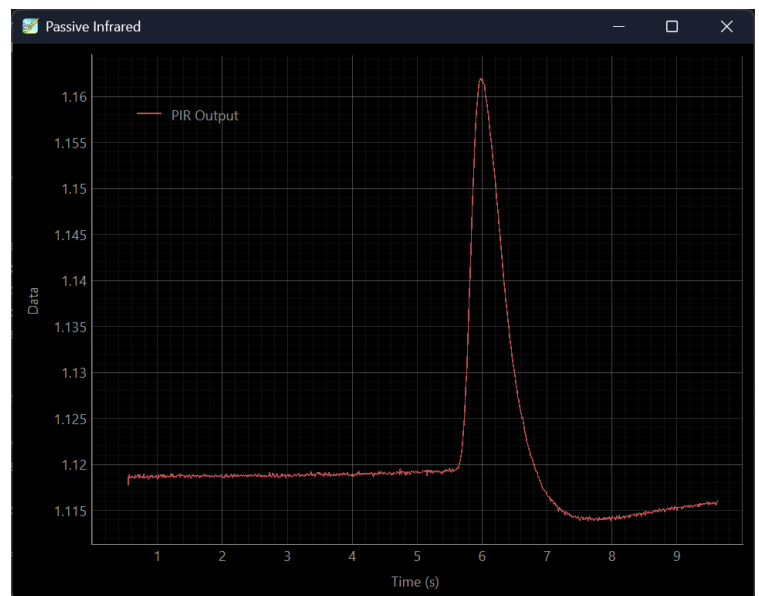


Figure 3: Representation of Passive Infrared sensor field of view

Sensor readings are measured in voltage and fluctuate depending if an object is moving within the PIR sensor's field of view. Figure 3 is a representation of the PL-Q873 comparing a static environment versus motion of an object crossing the field of view of the sensor.



a. PIR measurement of static environment



b. PIR measurement of motion detected

Figure 3. Example voltage measurement of a PIR sensor.

The rate at which the PIR sensor voltage rises and falls is related to the sensitivity of the ceramic sensor. Likewise, since PIR sensors measure infrared emission, the change in temperature between the ambient environment and the object present has an impact on how consistently a voltage spike can be detected.

© Quanser Inc., All rights reserved.



Solutions for teaching and research. Made in Canada.